Topic: Convergence of a Fourier series

Question: Find any points where the Fourier series representation doesn't converge to the periodic extension of $f(x)$.

$$f(x) = \begin{cases} (x+1)^2 & -L \leq x < 0 \\ (x-1)^2 & 0 \leq x \leq L \end{cases}$$

Answer choices:

A The function converges at every point

B $x = 2Ln$

C $x = L + 2Ln$

D $x = Ln$

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Solution: A

The function $f(x)$ is continuous and smooth on intervals $(-L,0)$ and $(0,L)$, so its Fourier series representation converges to $f(x)$ on these intervals, and we only need to check convergence at $x = 0$ and $x = L$.

At $x = 0$,

$$f(x) = (0 - 1)^2 = 1$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (x + 1)^2 = 1$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (x - 1)^2 = 1$$

Because these three values are equivalent, $f(x)$ is continuous at $x = 0$ and its Fourier representation converges to $f(x)$ at $x = 0$.

At $x = L$,

$$f(L) = (L - 1)^2$$

$$f(-L) = (-L + 1)^2 = (L - 1)^2$$

Therefore, $f(L) = f(-L)$, and the periodic extension of $f(x)$ is continuous at $x = L$. Then the Fourier series representation of $f(x)$ converges to $f(x)$ at $x = L$ and $x = -L$. Therefore, the Fourier series converges to the periodic extension of $f(x)$ at every point.

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Topic: Convergence of a Fourier series

Question: Find any points where the Fourier series representation doesn't converge to the periodic extension of $f(x)$.

$$f(x) = \begin{cases} \arctan\left(\frac{1}{x}\right) & x \in [-L,0) \cup (0,L] \\ \frac{\pi}{2} & x = 0 \end{cases}$$

Answer choices:

A The function converges at every point

B $x = 2Ln$

C $x = L + 2Ln$

D $x = Ln$

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Solution: D

The function $f(x)$ is continuous and smooth on the intervals $(-L,0)$ and $(L,0)$, so its Fourier series representation converges to $f(x)$ in these intervals, and we only need to check convergence at $x = 0$ and $x = \pm L$.

At $x = 0$,

$$f(x) = \frac{\pi}{2}$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \arctan\left(\frac{1}{x}\right) = -\frac{\pi}{2}$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \arctan\left(\frac{1}{x}\right) = \frac{\pi}{2}$$

Because the one-sided limits aren't equivalent, the function has a jump discontinuity at $x = 0$, and the Fourier series representation will converge to the average of the one-sided limits,

$$\frac{-\frac{\pi}{2} + \frac{\pi}{2}}{2} = 0$$

At $x = \pm L$,

$$\lim_{x \to -L} f(x) = f(-L) = \arctan\left(\frac{1}{-L}\right) = -\arctan\left(\frac{1}{L}\right)$$

$$\lim_{x \to L} f(x) = f(L) = \arctan\left(\frac{1}{L}\right)$$

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Because the one-sided limits aren't equivalent, the function has a jump discontinuity at $x = L$, and the Fourier series representation will converge to the average of the one-sided limits,

$$\frac{-\arctan\left(\frac{1}{L}\right) + \arctan\left(\frac{1}{L}\right)}{2} = 0$$

Therefore, the Fourier series representation will not converge to the periodic extension of $f(x)$ at every $x = 0 + 2Ln$ or at every $x = L + 2Ln$. We can express these together as every $x = Ln$.

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Topic: Convergence of a Fourier series

Question: Find any points where the Fourier series representation doesn't converge to the periodic extension of $f(x)$.

$$f(x) = \begin{cases} 2 + x^2 & x \in [-L, 0) \\ 1 & x = 0 \\ x^2 & x \in (0, L] \end{cases}$$

Answer choices:

A The function converges at every point

B $x = 2Ln$

C $x = (2n + 1)L$

D $x = Ln$

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Solution: C

The function $f(x)$ is continuous and smooth on the intervals $(-L,0)$ and $(L,0)$, so its Fourier series representation converges to $f(x)$ at in these intervals, and we only need to check convergence at $x = 0$ and $x = L$.

At $x = 0$,

$$f(x) = 1$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} (2 + x^2) = 2$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2 = 0$$

Because the one-sided limits aren't equivalent, the function has a jump discontinuity at $x = 0$, and the Fourier series representation will converge to the average of the one-sided limits,

$$\frac{2 + 0}{2} = 1 = f(x)$$

So the Fourier series actually converges to the value of $f(x)$ at $x = 0$.

At $x = L$,

$$\lim_{x \to -L} f(x) = f(-L) = 2 + L^2$$

$$\lim_{x \to L} f(x) = f(L) = L^2$$

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Because the one-sided limits aren't equivalent, the function has a jump discontinuity at $x = L$, and the Fourier series representation will converge to the average of the one-sided limits,

$$\frac{2 + L^2 + L^2}{2} = 1 + L^2 = f(L)$$

Therefore, the Fourier series representation will not converge to the periodic extension of $f(x)$ at every

$$x = L + 2Ln$$

$$x = (2n + 1)L$$

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